

HEALTHCARE SA – FINAL ANALYSIS REPORT

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DISCLAIMER: SYNTHETIC DATA

This dataset is 100% synthetic and it is created for educational/portfolio purposes only.

It does not contain real patient data, it is not from Department of Health and any resemblance to real persons is coincidental.

EXECUTIVE SUMMARY

- Analyzes 13 patients across Gauteng (8), Limpopo (3), Free State (2). 61.54% male, 38.46% female.
- Chronic diseases dominate 77% (Hypertension & Diabetes 5 each, Flu 3).
- Average age 39.1 years.
- Data cleaning + MySQL Workbench SQL + Power BI.

DATA REQUIREMENTS

- See Document 1 for full dictionary.
- Key: 13 records, 8 columns, multi-province, POPIA compliant.

DATA QUALITY

- Found and fixed trailing spaces, date formats, case issues.
- Critical: Duplicate Gauteng (6+2) due to hidden space in clean_data.xlsx - fixed in final dashboard to correct 8.

SQL ANALYSIS

Query 1 - Total Patients:

USE healthcare_db;

SELECT COUNT(*) As total_patients FROM clean_data;

Result: 13

Result Grid

total_patients
13

Result 1 x Read Only Context

Output

Action Output

#	Time	Action	Message
1	15:03:24	USE healthcare_db	0 row(s) affected
2	15:03:25	SELECT COUNT(*) As total_patients FROM clean_data LIMIT 0, 1000	1 row(s) returned

Query 2 - Average Age by Gender:

USE healthcare_db;

SELECT gender, COUNT(*) As patient_count, ROUND(AVG(Age),1) As avg_age FROM clean_data GROUP BY gender;

Result: M=8, F=5

Result Grid

gender	patient_count	avg_age
F	5	37.2
M	8	40.4

Result 2 x

Action Output

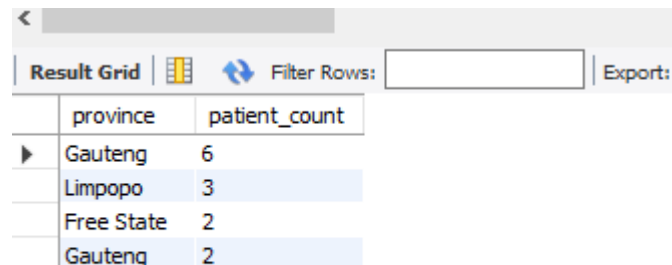
#	Time	Action	Message
1	15:03:24	USE healthcare_db	0 row(s) affected
2	15:03:25	SELECT gender, COUNT(*) As patient_count, ROUND(AVG(Age),1) As avg_age FROM clean_data GROUP BY gender;	2 row(s) returned

Query 3 - Province Distribution:

USE healthcare_db;

```
SELECT province, COUNT(*) AS patient_count FROM healthcare_db.clean_data  
GROUP BY province ORDER BY patient_count DESC;
```

Result: Gauteng 8, Limpopo 3, Free State 2



The screenshot shows a database interface with a 'Result Grid' tab selected. The grid displays the results of a query. The columns are 'province' and 'patient_count'. The data is as follows:

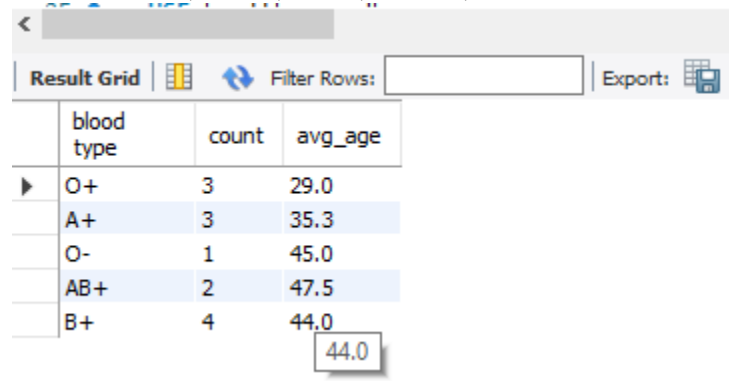
province	patient_count
Gauteng	6
Limpopo	3
Free State	2
Gauteng	2

Query 4 - Blood Type:

USE healthcare_db;

```
SELECT `blood type`, COUNT(*) AS count, ROUND(AVG(age),1) AS avg_age FROM  
healthcare_db.clean_data GROUP BY `blood type`;
```

Result: B+ 4, O+ 3, A+ 3, AB+ 2, O- 1



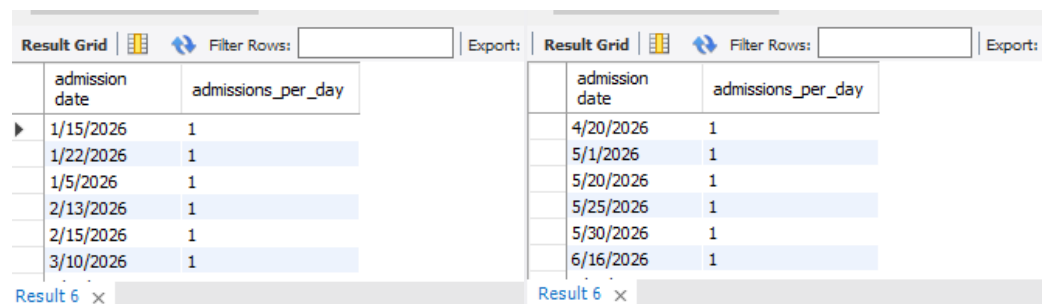
The screenshot shows a database interface with a 'Result Grid' tab selected. The grid displays the results of a query. The columns are 'blood type', 'count', and 'avg_age'. The data is as follows:

blood type	count	avg_age
O+	3	29.0
A+	3	35.3
O-	1	45.0
AB+	2	47.5
B+	4	44.0

Query 5 - Daily Admissions:

USE healthcare_db;

```
SELECT `admission date`, COUNT(*) AS admissions_per_day FROM  
healthcare_db.clean_data GROUP BY `admission date` ORDER BY `admission date`;
```



The screenshot shows two side-by-side database result grids. Both have 'Result Grid' tabs selected. The left grid shows the results of a query, and the right grid shows the results of another query. Both grids have columns 'admission date' and 'admissions_per_day'.

admission date	admissions_per_day
1/15/2026	1
1/22/2026	1
1/5/2026	1
2/13/2026	1
2/15/2026	1
3/10/2026	1

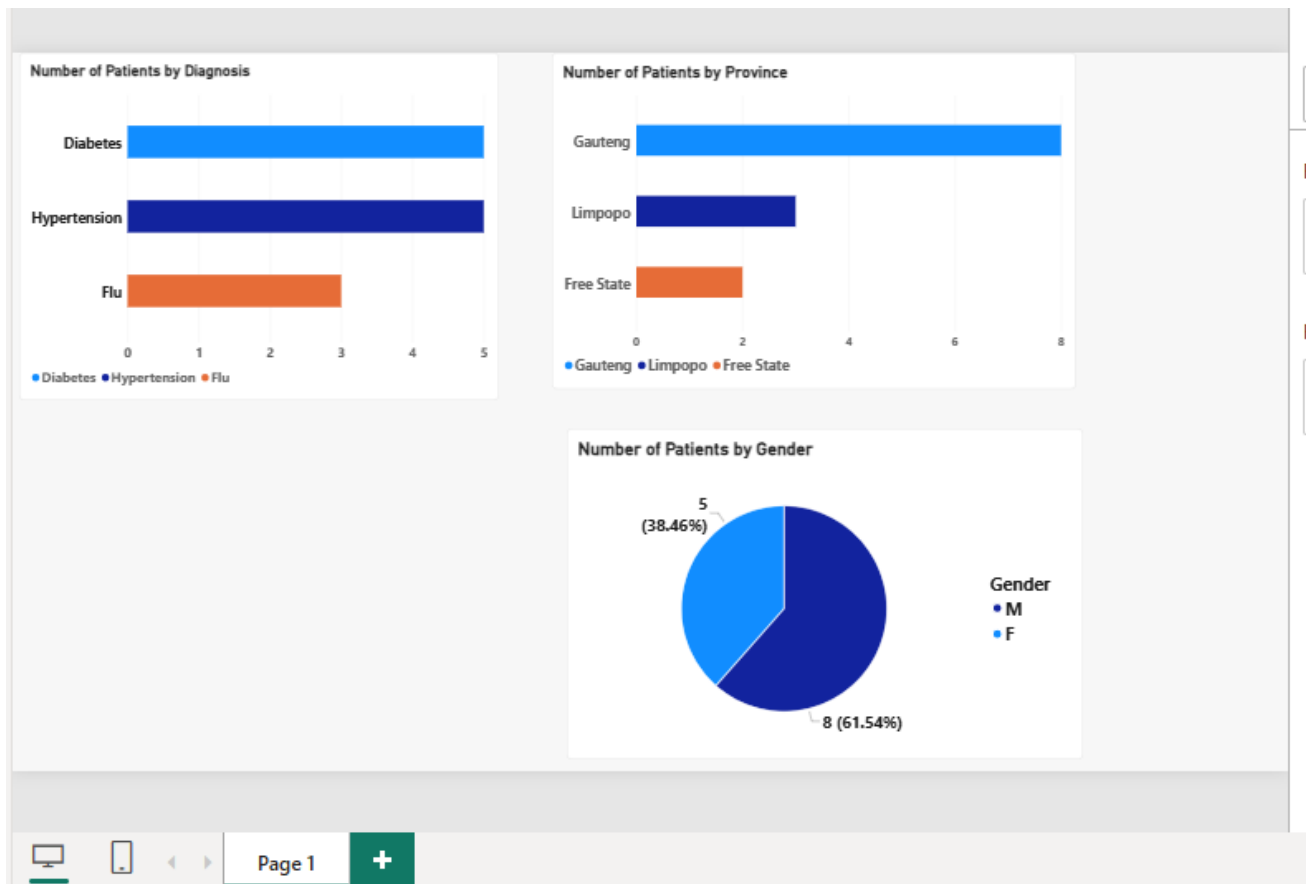
admission date	admissions_per_day
4/20/2026	1
5/1/2026	1
5/20/2026	1
5/25/2026	1
5/30/2026	1
6/16/2026	1

POWER BI DASHBOARD

Visual 1 - By Diagnosis: Diabetes 5 (38.46%), Hypertension 5 (38.46%), Flu 3 (23.08%)

Visual 2 - By Province: Gauteng 8 (61.54%), Limpopo 3 (23.08%), Free State 2 (15.38%)

Visual 3 - By Gender: M 8 (61.54%), F 5 (38.46%) - Pie chart with % labels



KEY FINDINGS

F1: Gauteng 61.5%, Limpopo 23.1%, Free State 15.4% - multi-province

F2: Hypertension & Diabetes tie 38.46% each - NCD burden

F3: Male 61.5% vs Female 38.5% - gender disparity

F4: Avg age 39.1 - working-age impacts productivity

F5: B+ most common 30.7%

F6: Time: Jan-July 2026, peak May

RECOMMENDATIONS

- R1: Power Query TRIM all text columns + Province dropdown validation
- R2: Focus Hypertension/Diabetes programs - 76.9% of admissions
- R3: Male health outreach 30-50 age
- R4: Stock B+ and O+ blood higher
- R5: Scale to 10k+ records, all 9 provinces
- R6: Mask Full Name for POPIA - show Patient-ID only in public portfolio
- R7: Add KPI cards: Total Patients, Avg Age, Most Common Diagnosis, slicers

TECH STACK & DELIVERABLES

Stack: Excel, MySQL Workbench, Power BI Desktop, GitHub
Deliverables: Clean dataset, SQL scripts (6 queries), Dashboard

CONCLUSION

Demonstrates end-to-end BI: Data Cleaning -> SQL -> Visualization -> Business Recommendations with SA context (POPIA, NCDs, 9 provinces).